



VANGUARD



A/B TEST ANALYSIS

Vanguard Digital UX Redesign

Did the New Interface Improve Client Outcomes?

June 12, 2026 | Ironhack DA Week 5 Bootcamp

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Project pipeline

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Experiment Overview

What was tested, who was involved, and how.

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Data & Cleaning

Sources, deduplication, and master dataset build.

03

Group Balance Check

Demographic parity between Control and Test.

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KPI Results — Diana

Error rates per funnel step + completion rates.

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KPI Results — Zidene

Process efficiency & step-duration analysis.

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Hypothesis Testing

Statistical significance of key findings.

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Conclusions & Recommendations

What the data says and next steps.

Experiment Overview

The two groups (Control/Test) ran the same 5-step process on different UIs. KEY question: did the redesign drove higher completion rates?.

PERIOD

March 15 – June 20, 2017

CONTROL

Legacy Vanguard UI

TEST

Redesigned UI + contextual prompts

CLIENTS

Control 23,527 · Test 26,961 ↑

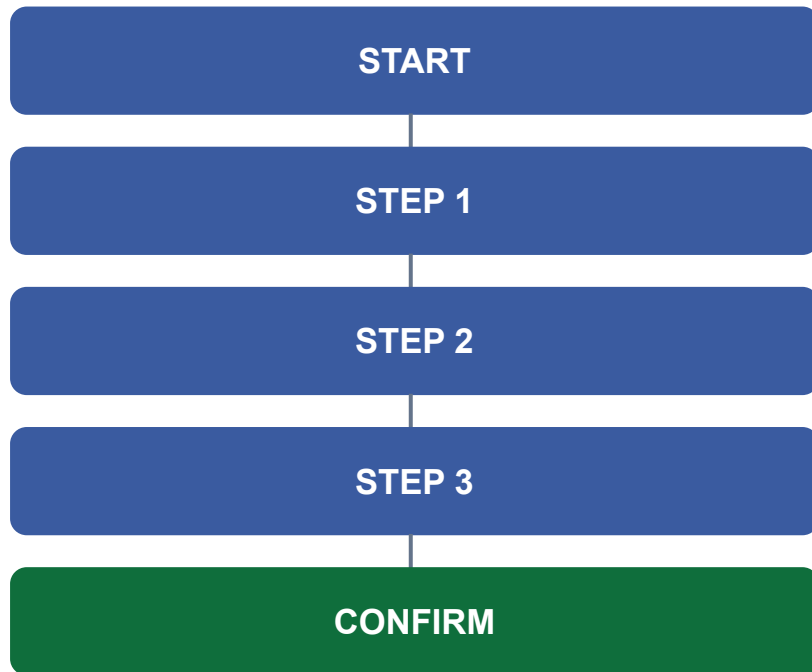
DATA ROWS

443,897 web interaction events

PRIMARY KPI

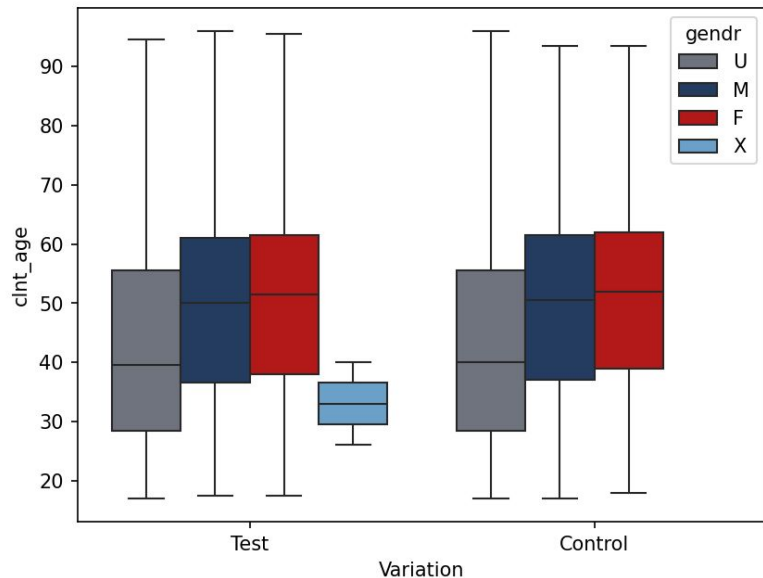
- Completion rate (client level)
- Process duration rate (whole process/ Step by step)

Process Flow



Group Balance Check

Before comparing KPIs we verify Control and Test are demographically equivalent.



Unknown segment is the largest gender group and skews ~8 yrs younger — included in KPIs.

Mean Age

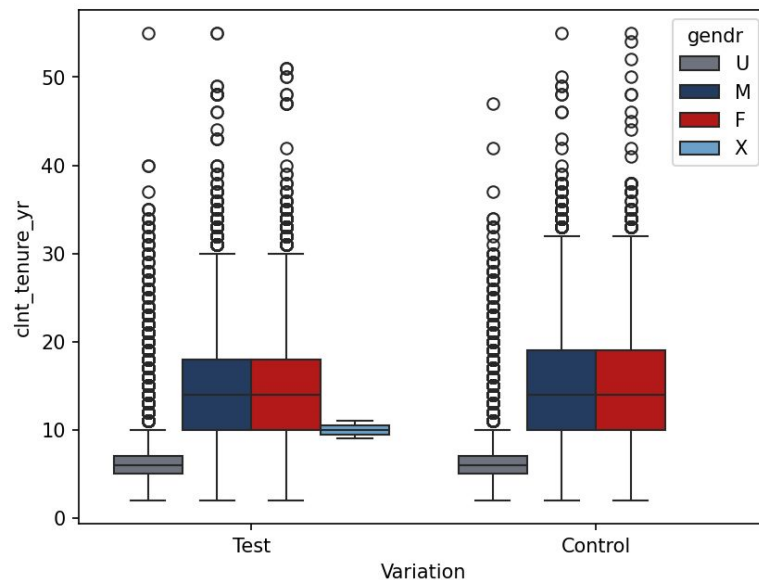
Control: 47.50 yrs
Test: 47.16 yrs

✓ Balanced

Age Std Dev

Control: 15.52 yrs
Test: 15.51 yrs

✓ Balanced



Mean Tenure

Control: 12.09 yrs
Test: 11.98 yrs

✓ Balanced

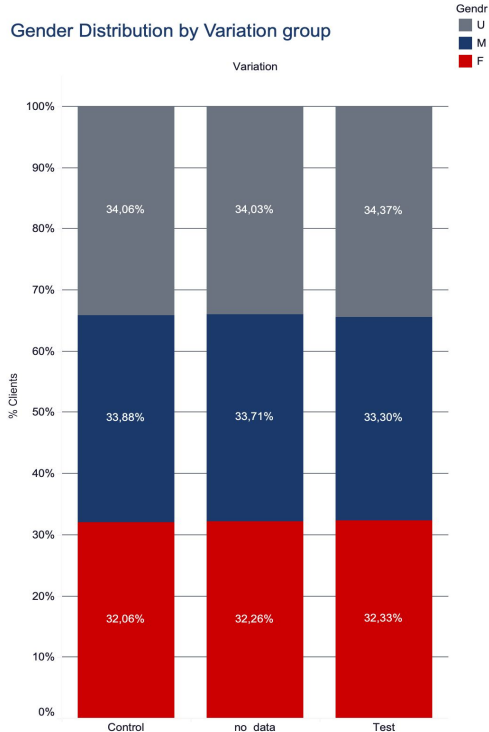
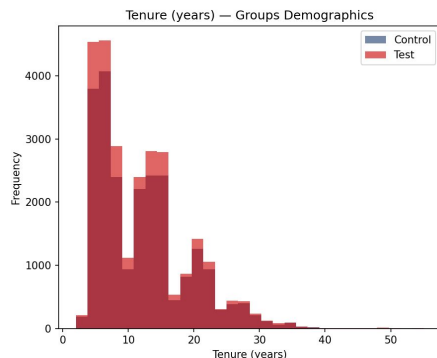
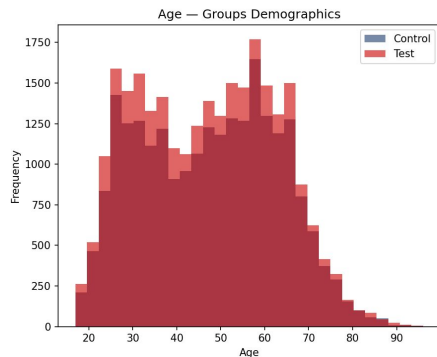
Tenure Std Dev

Control: 6.88 yrs
Test: 6.84 yrs

✓ Balanced

Group Balance Check

Before comparing KPIs we verify Control and Test are demographically equivalent.



Gender Distribution

Female ~32% · Male ~33% · Unknown (U) ~34% · X <0.01%
— near-identical across groups.
Clients with gender='X' **excluded from KPI analysis** (negligible sample).

✓ **Balanced**

CONCLUSION:

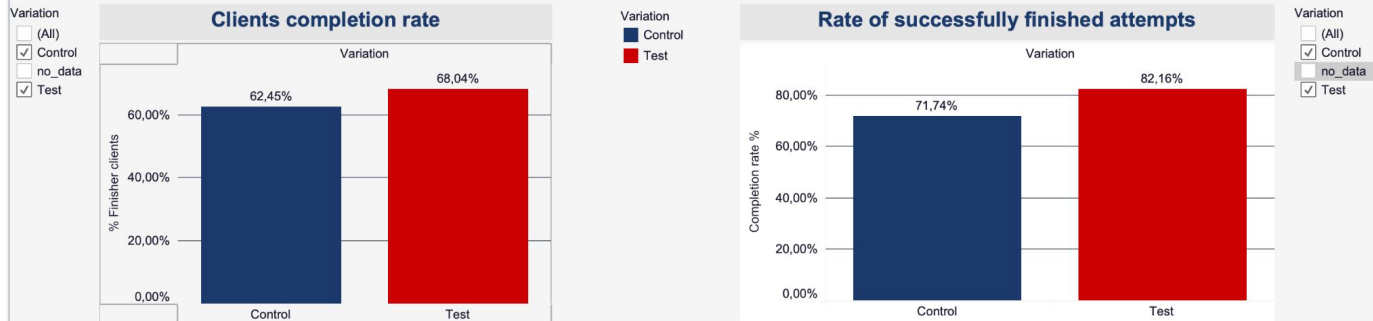
Both groups are **statistically comparable** — any KPI differences can be attributed to the UI redesign, not to group composition.

Error & Completion Rates

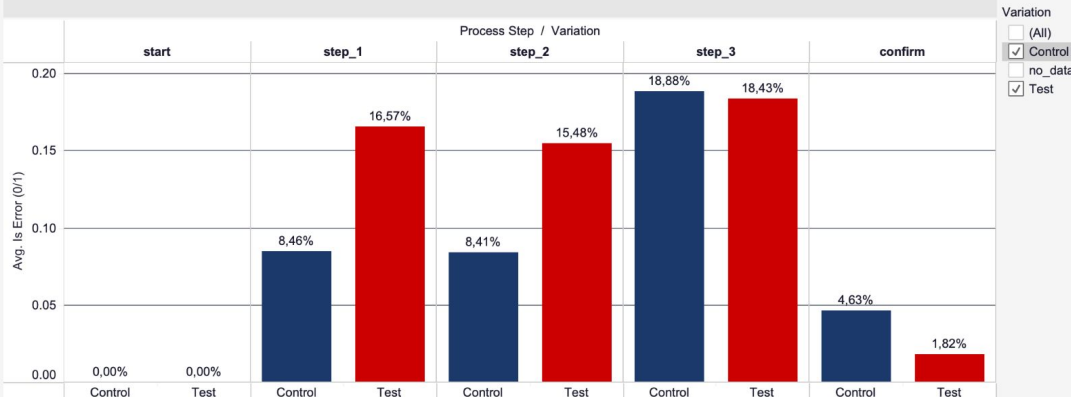
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Tableau Dashboard](#)



KPIs Summary



Error Rate per Funnel Step (%)



Key Insights

- New UI seems to increase **completion rate +5.6%**.
- Redesign causes more exploration mid-funnel (Steps 1–2).
- The decisive Confirm step, Test drops 1.82% error vs 4.63% for Control.

Net result: higher overall completion despite more mid-flow friction.

Hypothesis Testing



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Independent groups, questions on whether a proportion of clients finished/error or not -> One-sided z-tests for proportions.

H1 — Completion Rate

H_0 : completion rate Test group $\leq$ completion rate Control group (the UX change had no positive effect).
 H_1 : The completion rate of Test $>$ completion rate of Control

Metric: %_finisher_clients

Denominator: Unique clients per group

Conclusion:

z-stat: 13.1564

p-value: 0.0000

p_value is < 0.05 -> H_0 is rejected: completion rate of Test group is greater as the completion rate of Control group.

H2 — Error Rate

H_0 : The error ratio in Test group is $\leq$ Control group
 H_1 : The error ratio in Test group is $>$ Control group

Metric: mean backward-nav rate

Denominator: Steps 1–4 per group

Conclusion:

z-stat: 23.7421

p-value: 0.0000

p_value is < 0.05 , the H_0 is rejected, meaning that the error ratio in test group is $>$ than control group

Conversion/Drop Off Rate Comparison: Test vs Control Group

- Drop Off rate Control users higher 34% (30% Test users)

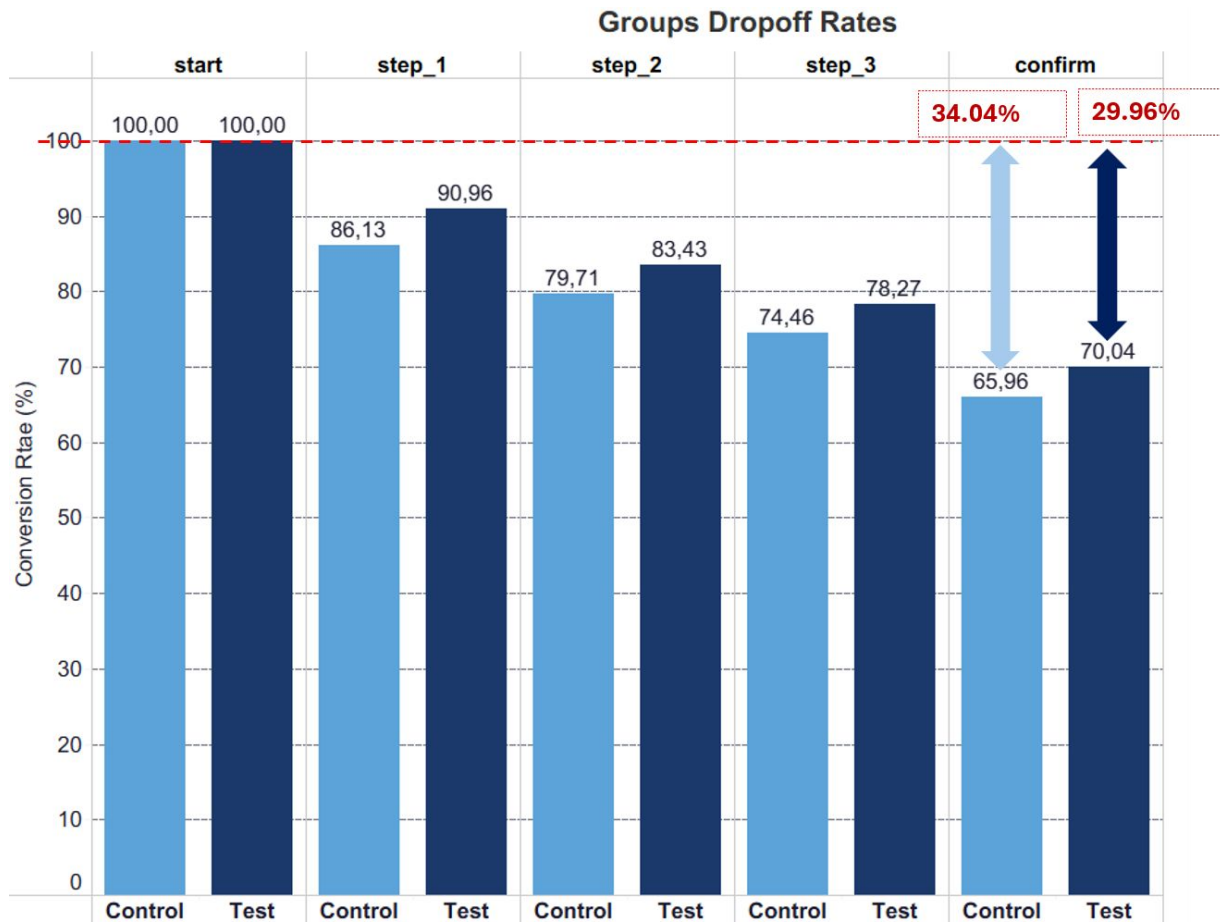
- Where do users most drop out?

Control users:

Step_1 = ~14%

Test users:

Step_1 = ~9%



Average Distribution of the Completion Time

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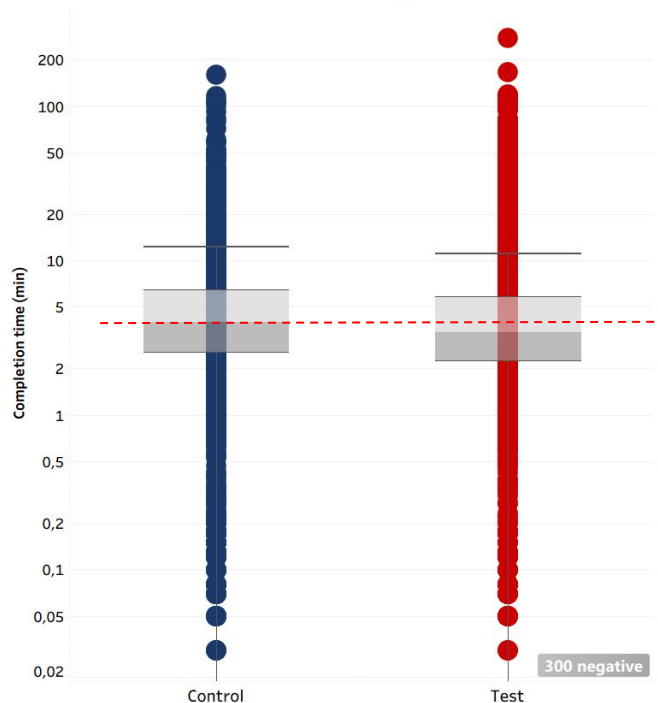


✓ Balanced

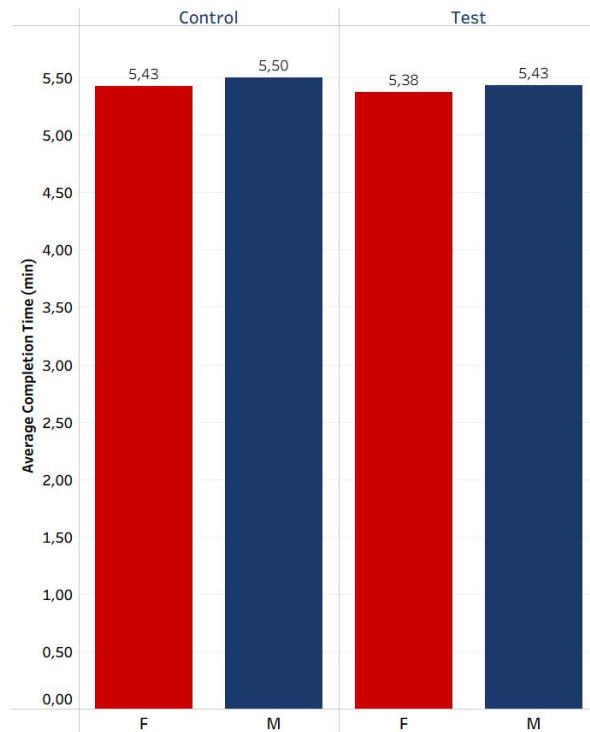
- Test Group is slightly faster in completing all steps, though difference is not significantly higher
- Males and females have nearly the same average of completion time.

Average Distribution of the Completion Time

Log Distribution of the Completion Time



Av. Completion Time by Group by Gender

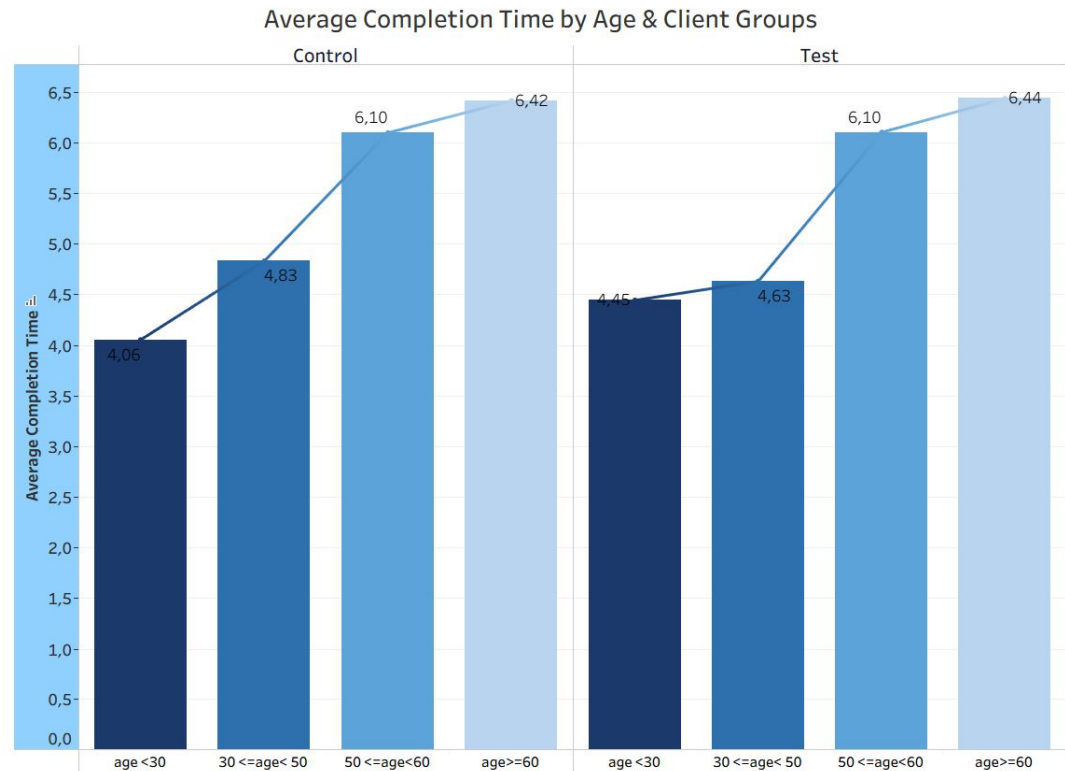


Average Completion Time by Age Segments

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✓ Balanced



Both groups follow the same trend in completion time for different age segments.

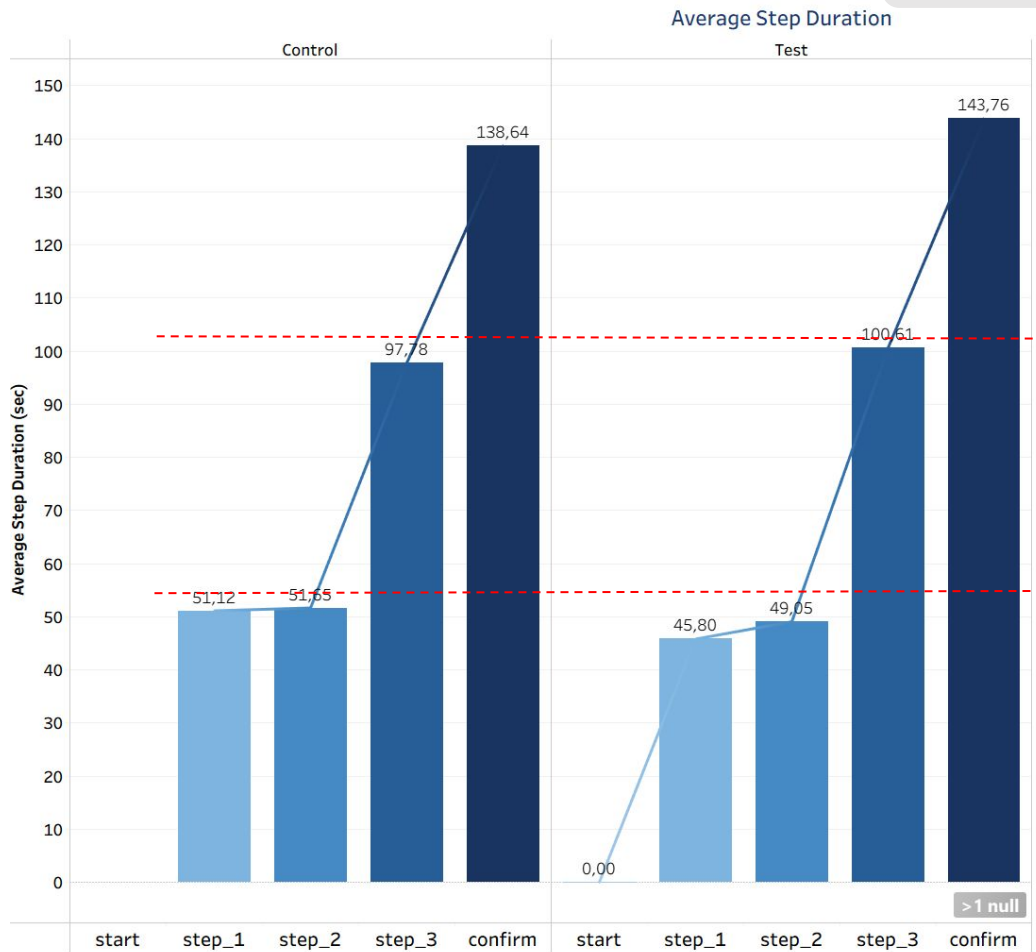
Step by Step Duration Comparison

✓ Balanced

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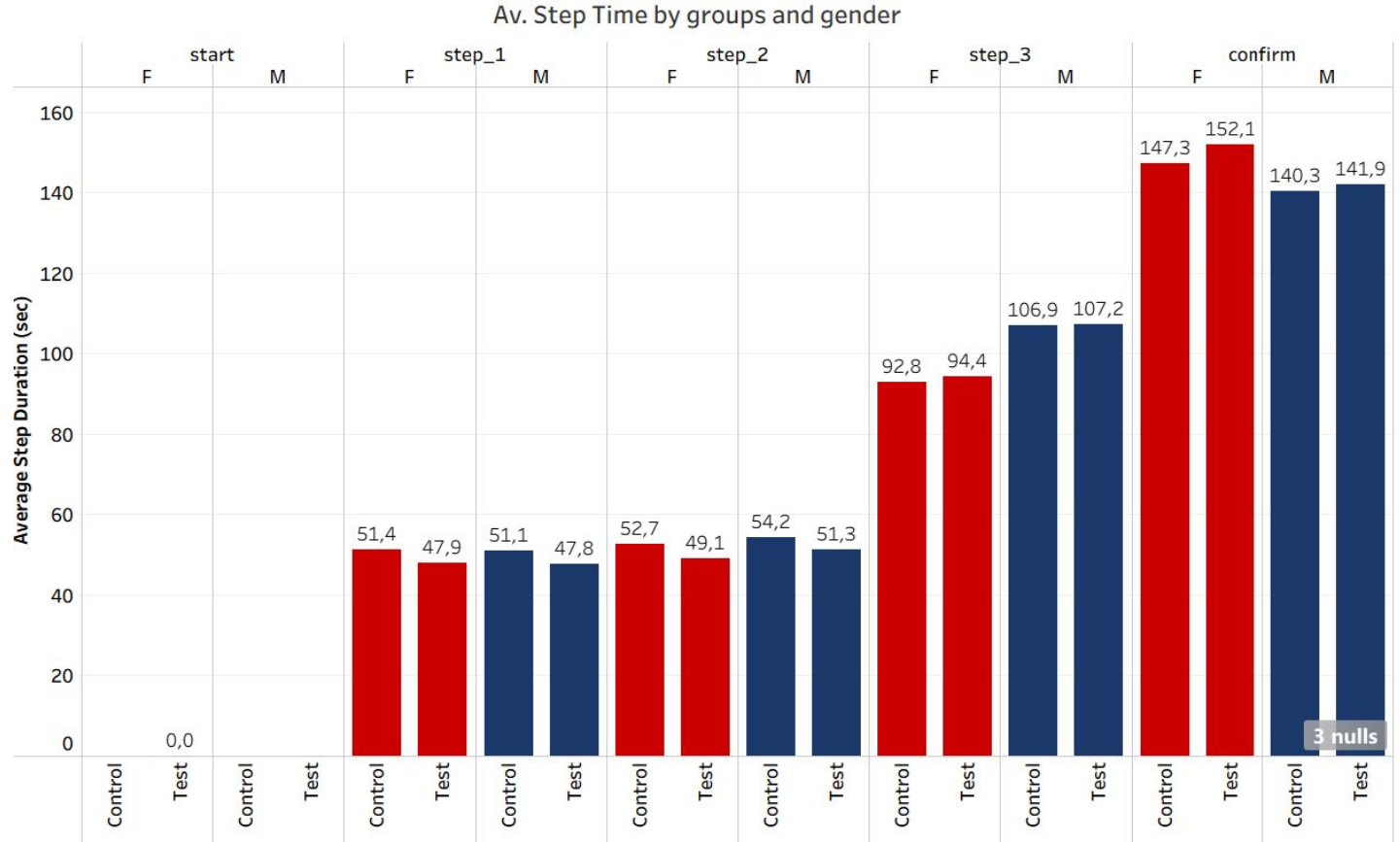


Same tendency for the two groups.
Last steps need more time for
validation.



Average Step Time by User Group and Gender

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Homogeneity between Males & Females in the step time for both groups:

- First 2 steps: around same mean time.

- Third step: females did better, however in the last step, males caught up

Hypothesis Testing



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H3 — Completion time

H₀: test_duration_mean = control_duration_mean

H₁: test_duration_mean < control_duration_mean

Metric: Mean completion time

Test: Two-sample t-test

Conclusion: Fail to reject H₀ (p_value = 0.37). There is insufficient evidence that the mean duration of the Test group is lower than that of the Control group.

Conclusions & Recommendations

Did the New Interface Improve Client Outcomes?

Better user experience?

🎯 Cleaner Confirm Step

Test drops confirm-step errors to 1.82% vs 4.63%. Users commit more confidently at the decisive final step.

⚠️ Same Process Speed

Mainly, the difference in step completion and overall completion is insignificant.

⚠️ Higher backtrack

Steps 1–2 error rates nearly double in Test (~16% vs ~8%). Users explore/backtrack more with the new interface.



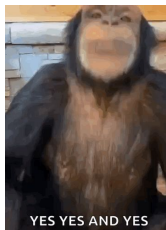
Higher process completion rates?

👤 Process completion effectiveness

Test users have better effectiveness than Control one. Drop rate Test group (30%) is less than Control group (34%)

✅ New UI Lifts Completion

Test clients complete the process +5.6 pp than Control group. Per-attempt rate rises to 82.2% vs 71.7%.



Recommendations

🚀 DEPLOY

Completion gain (+3.6 pp); statistical significance confirmed.

🔍 INVESTIGATE

High error rates on Steps 1–2 in Test: A/B test UX copy and prompts to reduce friction.



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Thank You

Questions?

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Ironhack Data Analytics Bootcamp

